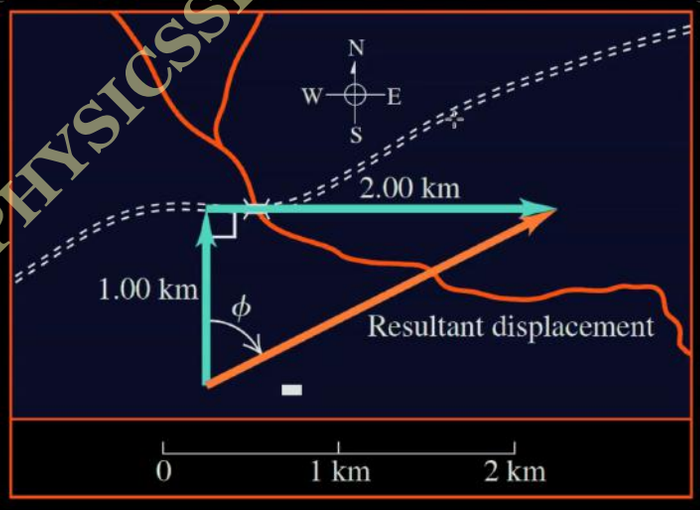
VECTOR: A quantity which has a magnitude as well as direction and gives a meaningful quantity upon vector addition rule is a vector. For instance displacement, Force, velocity, acceleration and momentum are some of the vector quantities. The physical quantities which can be represented by a single numerical value without any reference to any direction is called a scalar. For instance charge, speed, time, mass and length are scalars.

→ A vector in space generally required specification of 3 numerical quantities while scalar requires only one.

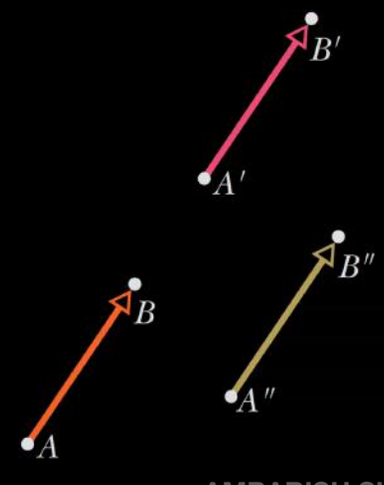
→ There are physical quantities which may require a 3×3 matrix for specification, such quantities are known as tensors (for instance Stress and strain are tensors).

REPRESENTATION OF A VECTOR: Just as maps are drawn to a certain scale, in the same manner vectors are also drawn to the scale. The vectors are drawn as arrows with lengths proportional to the physical quantity being represented and the direction same as that of physical quantity.



For instance in the shown figure, the displacement $2\,km\,\hat{E}$ has been shown with an arrow of double the length as used for representing the displacement $1\,Km\,\hat{N}$. ($\hat{E}$ and $\hat{N}$ are called unit vectors along East and North respectively).

EQUALITY OF TWO VECTORS: Two or more vectors are said to be equal if they have same magnitudes and directions irrespective of where they are located in space. Thus in the adjacent figure $\overrightarrow{AB}$, $\overrightarrow{A'B'}$ and $\overrightarrow{A''B''}$ are equal vectors though represented at different points of space.



→ Two vectors being equal does not necessarily mean that they have same physical effect.

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For instance a train starting at Chennai may have same velocity vector as another train starting at Delhi however they are passing through different points of space. Similarly equal forces applied at different distances from a door hinge can produce different rotational effects. (This is somewhat like two different people having equal amount of money, when we say that it only refers to the amounts and nothing more, their wealth's may have different impacts on their lives and society).

Notation For Vector Variables: A vector variable is generally represented by putting an arrow over an English or Greek alphabet symbol. For instance a force variable may be represented as $\vec{F}$.

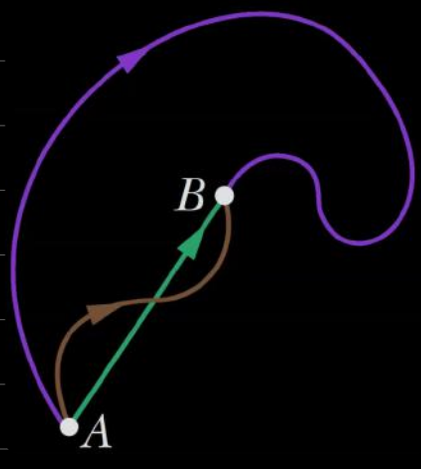
→ Some books may use alternative notation such as $\bar{F}$ or $\underline{F}$ or just a bold font **F** (Irodov's notation).

→ Unit vectors are represented by putting a hat over the alphabet symbols such as $\hat{N}$ can represent unit vector along North. (A unit vector has a magnitude 1, some direction and no units).

→ For every vector variable $\vec{F}$ and associated symbol $\hat{F}$ represents a unit vector along $\vec{F}$. For instance, if $\vec{F} = 2N\,\hat{E}$ then, $\hat{F} = 1\hat{E}$.

→ For every vector variable $\vec{F}$ an associated symbol F or $|\vec{F}|$ represents just the magnitude of the vector (always considered positive). Thus if $\vec{F} = N\,\hat{E}$ then $F = 2N$.

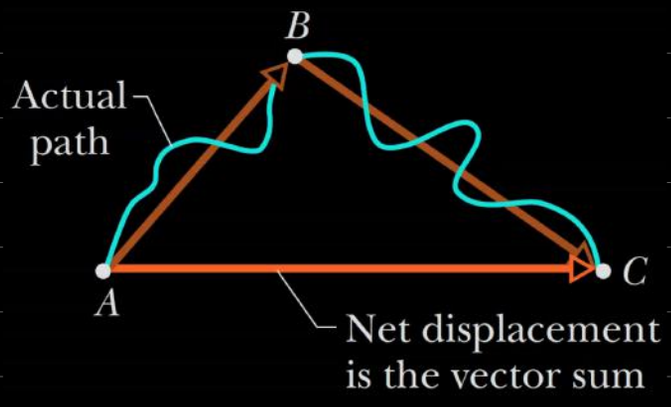
Displacement Vector: A vector from initial position of a particle to final position of a particle is called displacement vector. For instance in adjacent figure, a particle moves along the purple path to reach from A to B and the displacement is represented by the green colored vector $\vec{AB}$. Thus displacement vector is independent of the exact path followed.



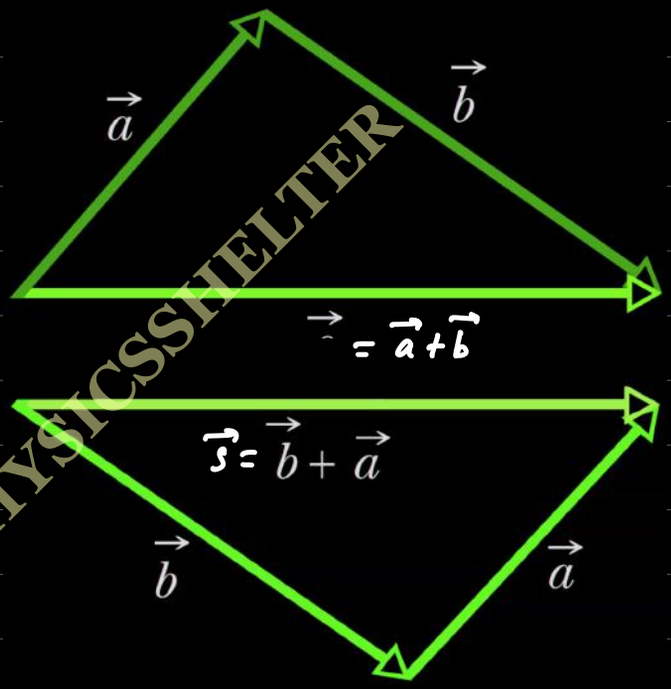
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VECTOR ADDITION:

Suppose a particle undergoes two successive displacements $\vec{AB}$ and $\vec{BC}$ then for finding the net displacement, we can use one of the following vector addition laws:



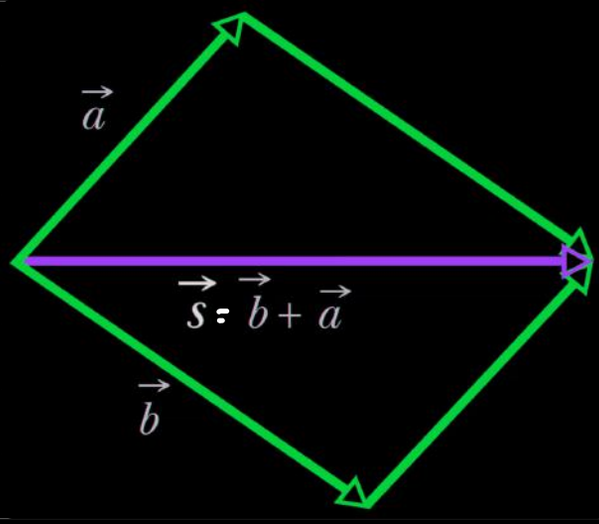
TRIANGLE LAW OF ADDITION: In this method, we arrange the vectors in a head to tail configuration (see adjacent figure), now net displacement is given by the vector joining tail of the first vector to the head of the second vector.



→ Note that for addition the order of arranging the vectors is not important (see adjacent figure). In first case we have drawn $\vec{b}$ from head of $\vec{a}$ while in second case we have drawn $\vec{a}$ from head of $\vec{b}$ but final resultant $\vec{s}$ is same, thus $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ or vector addition is commutative.

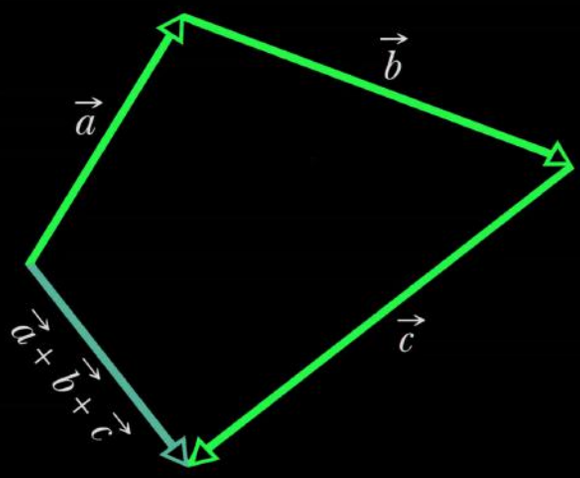
PARALLELOGRAM LAW OF ADDITION:

In this method, we arrange the vectors in a tail to tail configuration and then complete the parallelogram, Now the resultant vector is given by the diagonal of the parallelogram pointing away from common tail of vectors. One can readily recognize that parallelogram law is a direct corollary of triangle law of addition.

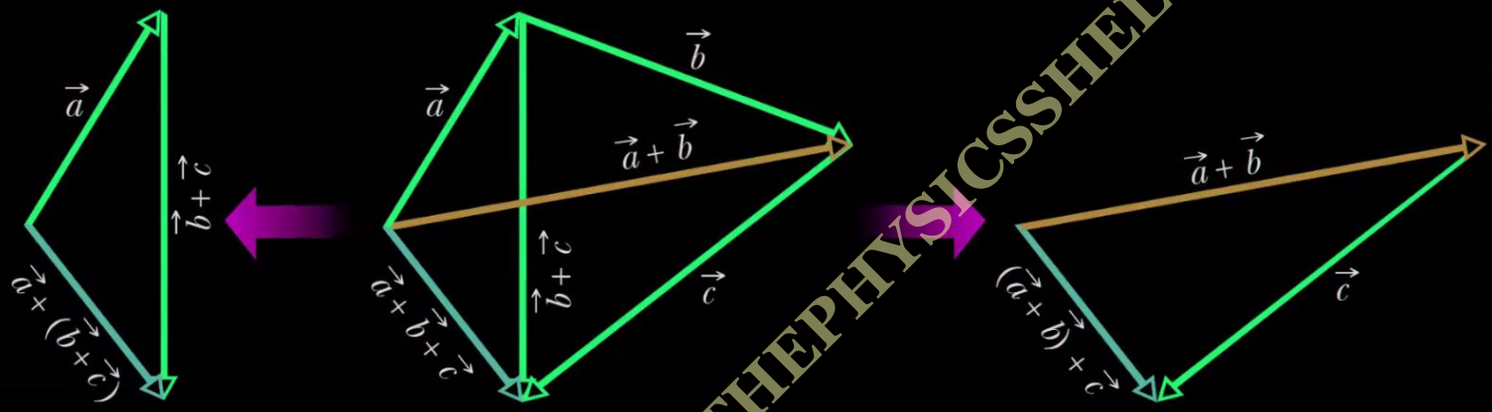


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Polygon Law Of Addition: For addition of more than two vectors, we can arrange all the vectors in a head to tail configuration and then the resultant vector is given as the vector joining the tail of the first vector and head of the last vector, as an example, consider the adjacent figure, here $\vec{a}+\vec{b}+\vec{c}$ is the closing side of the polygon defined by vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.



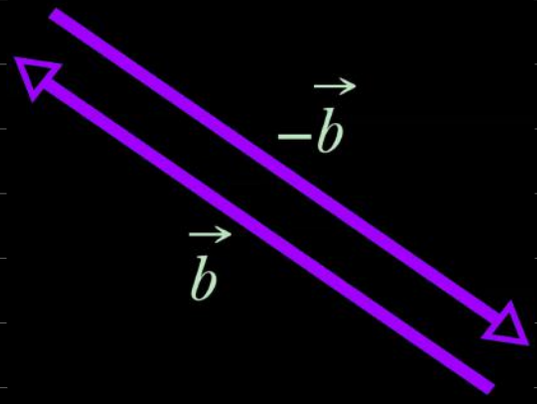
→ Polygon law of addition follows from triangle law of addition:



→ From above figure it is also clear that vector addition also follows associative property i.e $(\vec{a}+\vec{b})+\vec{c} = \vec{a}+(\vec{b}+\vec{c})$

Negative Of A Vector:

Negative of a vector is another vector whose magnitude is same but the direction exactly opposite (see adjacent figure). Thus if $\vec{F} = SN\hat{E}$ then $-\vec{F} = -SN\hat{E}$ or $SN\hat{W}$.

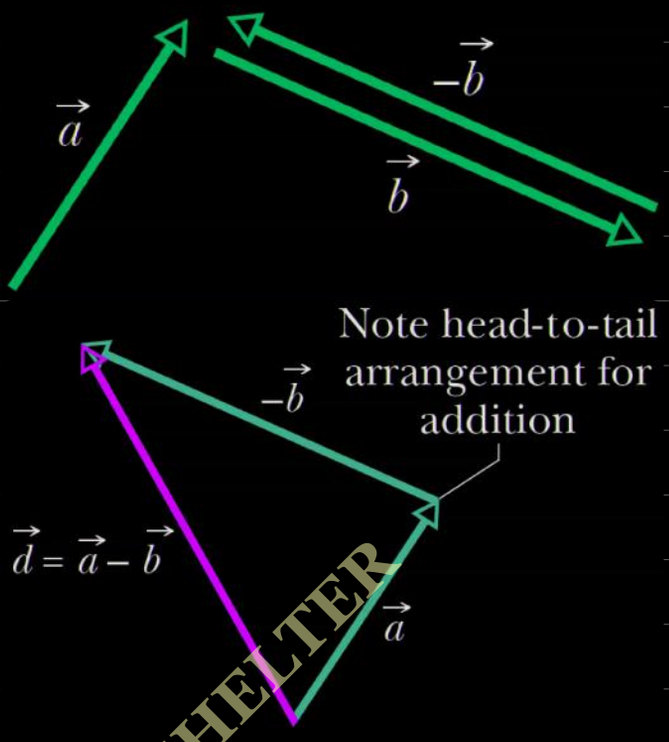


→ If $\vec{F} = -SN\hat{E}$ F is still SN and not $-SN$, recall that magnitude of a vector is always considered as positive.

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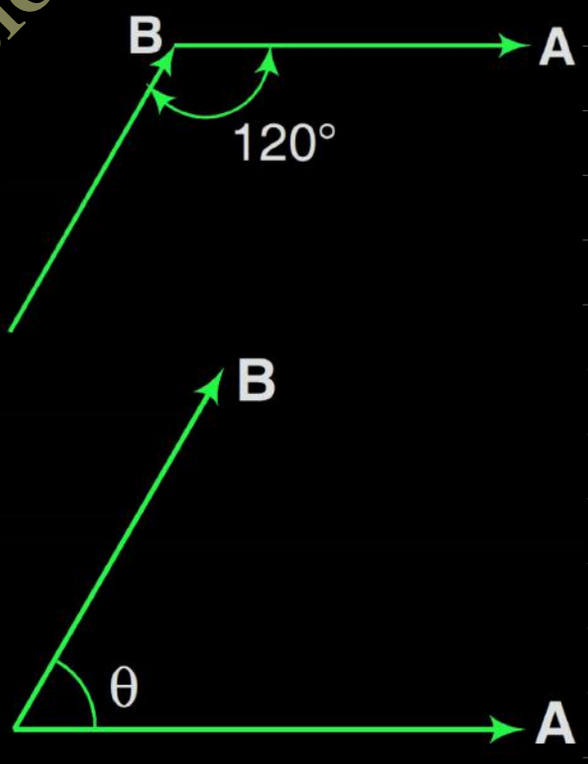
Vector Subtraction:

Subtraction of a vector is simply equivalent to addition of its negative for instance if we want to find $\vec{a}-\vec{b}$, then first we find $-\vec{b}$ and then perform the addition $\vec{a}+(-\vec{b})$ according to triangle law. See adjacent figure. (Remember to arrange vectors in head to tail config for triangle law). Alternatively we could use llgm law by arranging vectors tail to tail.



Angle Between Vectors:

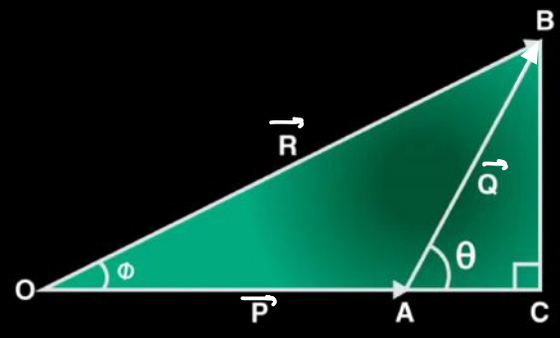
Angle between two vectors is defined as the non reflex angle between them when they are joined tail to tail (or head to head). For instance in the adjacent figure, angle between vectors is not 120° because they are in head to tail config. We need to arrange them as shown in the next figure. From this figure we can now infer that the angle between vectors $\vec{A}$ and $\vec{B}$ is $\theta = 180° - 120°$ i.e $\theta = 60°$. It is important to correctly work out the angle for analytic addition which we shall see in the next section.



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Analytical Addition Of Vectors:

Let $\vec{P}$ and $\vec{Q}$ be the two vectors being added by triangle law as shown in the figure. Note that angle between the vectors is θ and not $180°-\theta$. We are interested in finding the resultant vector $\vec{R}$ which is defined if we find angle ϕ and R.



Now using Pythagoras theorem:

$$R^2 = (P+Q\cos\theta)^2 + (Q\sin\theta)^2 = P^2 + 2PQ\cos\theta + Q^2\cos^2\theta + Q^2\sin^2\theta$$

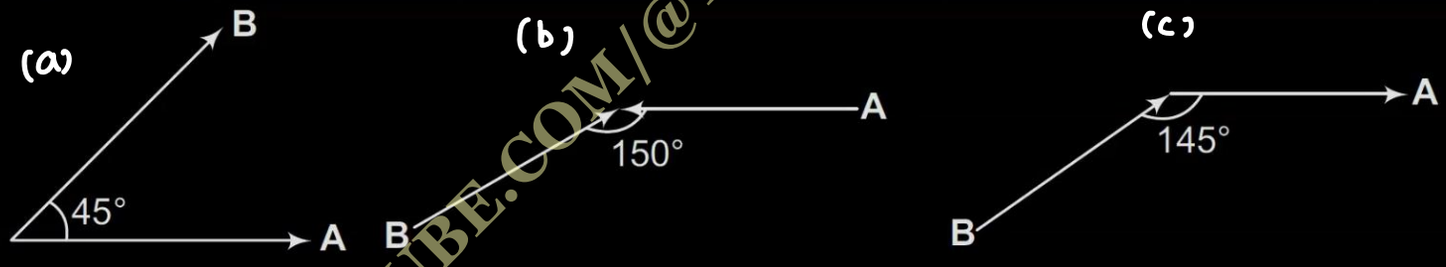
Thus:

$$\boxed{R^2 = P^2 + Q^2 + 2PQ\cos\theta}$$

Now $\tan\phi = BC/OC = Q\sin\theta/(P+Q\cos\theta)$ thus:

$$\boxed{\tan\phi = \frac{Q\sin\theta}{P+Q\cos\theta}}$$

Example:



What are the angles between shown vectors in the three cases.

Solution: (a) $45°$ (∵ already tail to tail)
(b) $150°$ (∵ already head to head)
(c) $180°-145° = 35°$ (∵ Vectors are in head to tail config.).

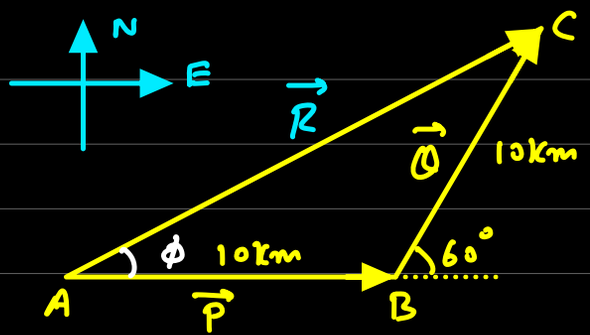
Example: A person travels 10 Km $\hat{E}$ followed by 10 Km at $60°$ North of east, find the displacement of the person.

Solution: First we shall represent the vectors in the triangle law of addition configuration and then perform analytic addition.

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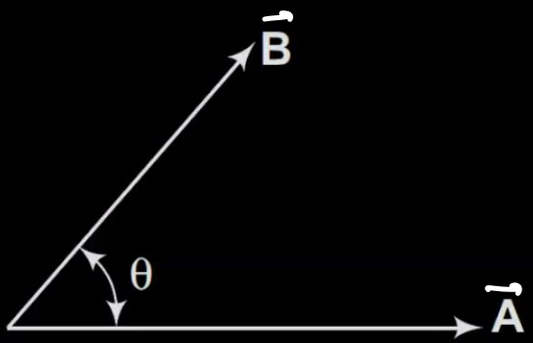
Consider the adjacent diagram.
$R^2 = P^2 + Q^2 + 2PQ\cos\theta$ thus:

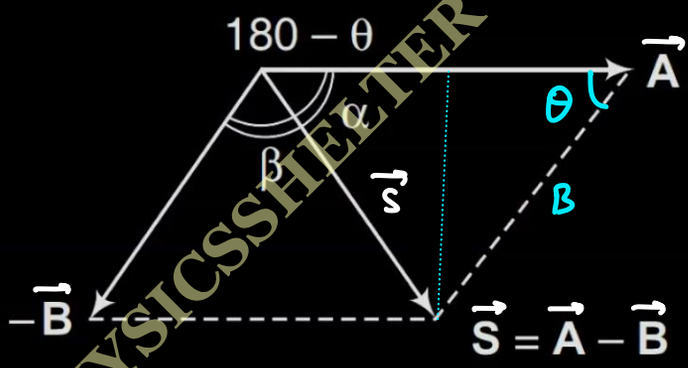
$R = \sqrt{100 + 100 + 200\cos 60^\circ}$ Km $= 10\sqrt{3}$ Km



Now $\tan\phi = 10\sin 60^\circ / (10 + 10\cos 60^\circ) = 1/\sqrt{3} \Rightarrow \phi = 30^\circ$.
Thus total displacement, $\vec{R} = 10\sqrt{3}$ Km @ 30° North of East.

Analytical Subtraction Of Vectors:





Suppose we have two vectors $\vec{A}$ & $\vec{B}$ and we wish to find $\vec{A} - \vec{B}$. If the angle between $\vec{A}$ & $\vec{B}$ is θ then angle between $\vec{A}$ and $-\vec{B}$ must be $180^\circ - \theta$. Thus according to the formula we derived for analytical addition:

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 + 2AB\cos(180^\circ - \theta)}$$

But $\cos(180^\circ - \theta) = -\cos\theta$ thus:

$$\boxed{|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB\cos\theta}}$$

We can find $\tan\alpha$ using geometrical construction (see turquoise vertical dotted line). $\tan\alpha = \dfrac{B\sin\theta}{A - B\cos\theta}$.

Multiplication Of A Vector By A scalar: When a vector is multiplied by a scalar, magnitudes and units get multiplied and the direction remains same if the scalar has positive value and gets reversed if scalar has negative value.

→ Division by a scalar is simply multiplication by its reciprocal.

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→ A vector divided by its own magnitude gives a unit vector along the vector, for instance if $\vec{F} = 20N\hat{E}$ then $F = 20N$ and $\vec{F}/F = 20N\hat{E}/20N = 1\hat{E}$ which is the unit vector along $\vec{F}$.

NULL VECTOR (ZERO VECTOR): A vector of zero magnitude is called a null vector or a zero vector. A null vector can be assigned any arbitrary direction ($0\,Km\,\hat{N}$ same as $0\,Km\,\hat{E}$ etc).

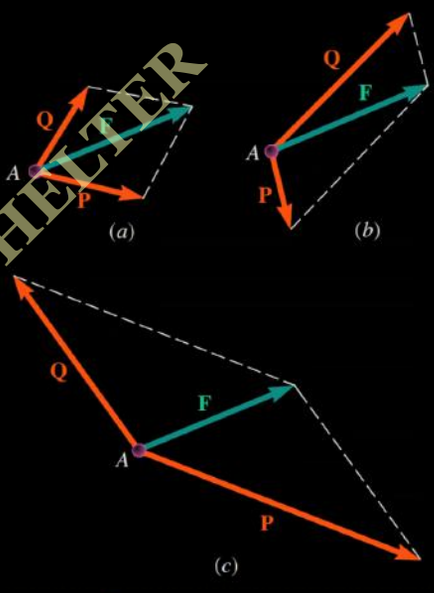
COMPONENTS OF A VECTOR:

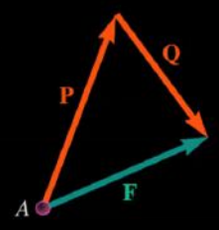
When a given vector is expressed as a sum of two vectors, the two vectors are said to components of the vector and the process is called resolving a vector in to its components. There can be infinite ways of resolving a vector in to components. For instance three ways of resolving $\vec{F}$ in to components $\vec{P}$ and $\vec{Q}$. (Think how many types of parallelograms are possible while maintaining $\vec{F}$ as its diagonal .... Infinite). In JEE Physics we are generally interested in components where $\vec{P} \perp \vec{Q}$ such components are called rectangular components.

→ Adjacent figure shows the process of resolving a vector in to components in two cases.

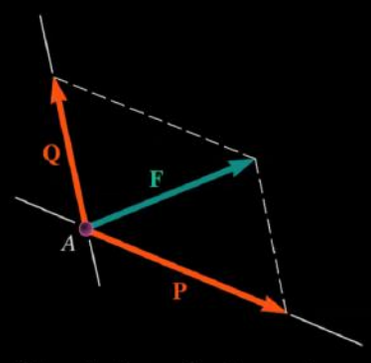
→ When one of the components is exactly known, we can use triangle law to find the other component, i.e if $\vec{P} + \vec{Q} = \vec{F}$ then $\vec{Q} = \vec{F} - \vec{P}$.

→ If directions of components are known, enclose $\vec{F}$ in a parallelogram with sides parallel to the directions of two forces.





When component P is known, use the triangle rule to find component Q.

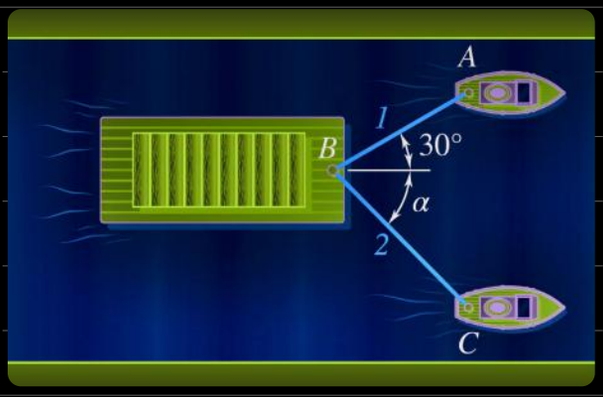


When the lines of action are known, use the parallelogram rule to determine components P and Q.

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Example: (Beer And Johnston):

Two tugboats are pulling a barge. If the resultant of the forces exerted by the tugboats is a 5000N force directed along the axis of the barge, determine (a) the tension (i.e., Pulling Force) in each of the ropes, given that $\alpha = 45°$, (b) the value of α for which the tension in rope 2 is a minimum.

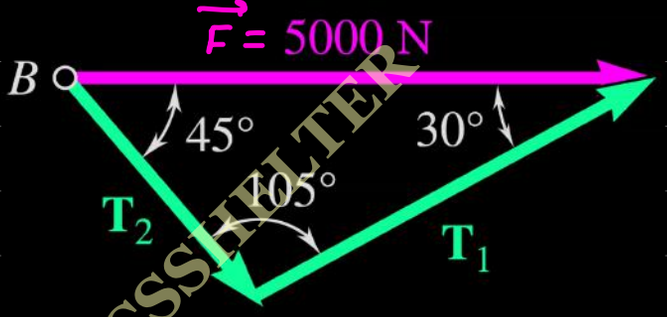


Solution: (a) Consider the vector diagram in adjacent figure. Applying Sine law:

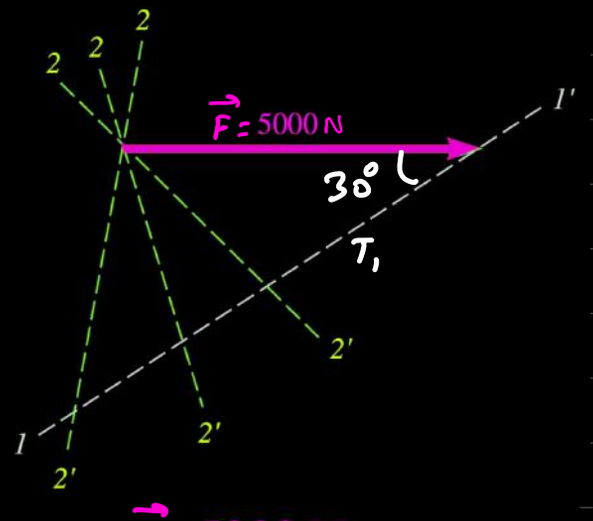
$$\frac{5000}{\sin 105°} = \frac{T_2}{\sin 30°} = \frac{T_1}{\sin 45°}$$

$$\Rightarrow T_1 = \frac{5000 \sin 45°}{\sin 105°} \approx 3660\ N$$

$$T_2 = \frac{5000 \sin 30°}{\sin(105°)} \approx 2590\ N \quad (\sin 105° = \sin(60° + 45°)).$$



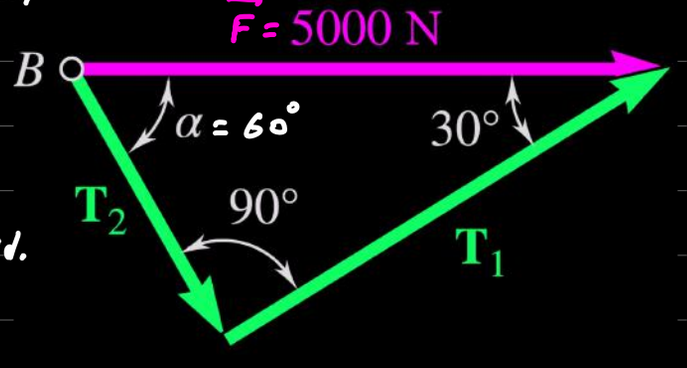
(b) Here line of action of T_1 is fixed though its magnitude can be varied so as to minimize T_2. Various possibilities of line of action are shown by green dotted lines. Also $\vec{T}_1 + \vec{T}_2$ must add up to $\vec{F}$. Clearly smallest value of T_2 occurs, when its line of action is chosen $\perp$ to line of action of T_1 (See adjacent figure).



Now, from the triangle diagram: $\alpha = 60°$ Ans. Corresponding value of T_2 and T_1 can also be calculated.

$T_2 = 5000 \cos 60° = 2500$ N and

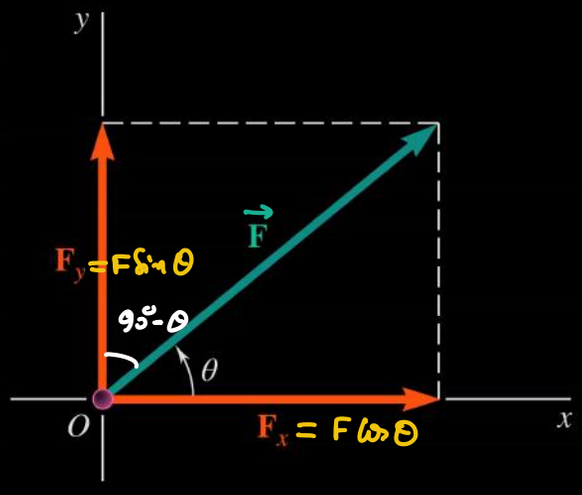
$T_1 = 5000 \cos 30° = 2500\sqrt{3}$ N Ans.



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Rectangular Components:

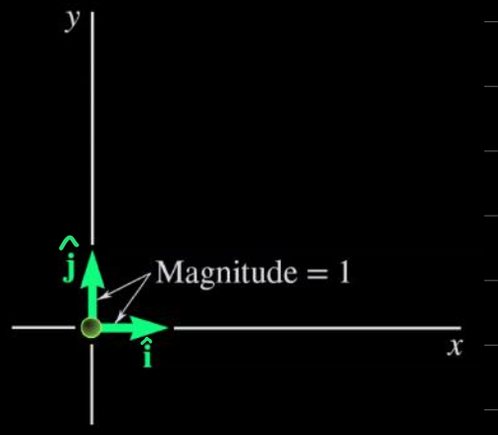
In our study of physics, very often we need to resolve a force in to components which are perpendicular to each other (see figure). For this purpose we complete a rectangle around the force and the components are given by the sides of rectangle. For instance in the adjacent figure, x component of force, $F_x = F\cos\theta$ and y component, $F_y = F\cos(90°-\theta) = F\sin\theta$. To get rectangular component of a force in any direction, we just need to multiply its magnitude by $\cos\theta$ where θ is the angle between the force and that direction.

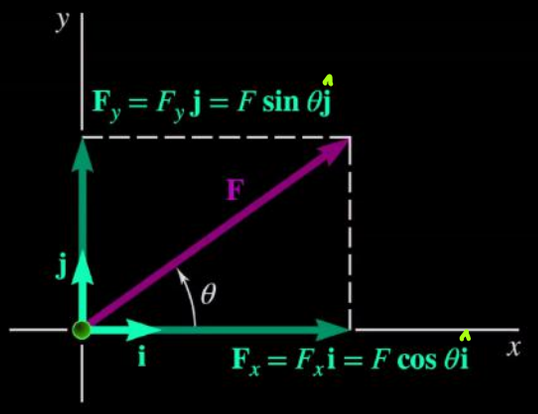


Rectangular Components Of A Force In Terms Of Unit Vectors:

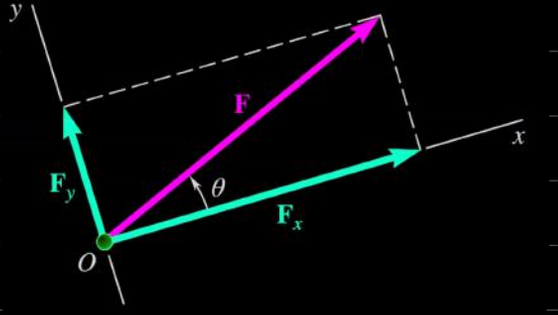
Recall that a unit vector is a vector of magnitude 1 and a certain direction. For sake of convenience of notation we define $\hat{i}$ as unit vector along x axis and $\hat{j}$ as unit vector along y axis. Having defined this now we can conveniently resolve a vector in it its $\hat{i}$ and $\hat{j}$ components. Suppose a force $\vec{F}$ makes an angle θ with +ve x-axis as shown, then, $F_x = F\cos\theta$ and $F_y = F\sin\theta$, combining the two components, we can write:

$$\vec{F} = F\cos\theta\,\hat{i} + F\sin\theta\,\hat{j}$$





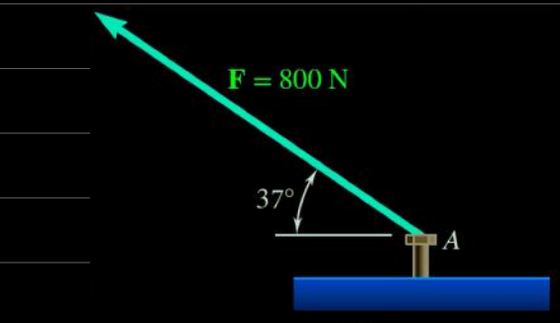
→ Note that x and y directions need not necessarily be along horizontal and vertical directions. Many times for sake of convenience, we might take inclined coordinate axes as shown. Here also, $\vec{F} = F\cos\theta\,\hat{i} + F\sin\theta\,\hat{j}$.



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→ $F\cos\theta\,\hat{i}$ and $F\sin\theta\,\hat{j}$ are said to be vector components while $F\cos\theta$ and $F\sin\theta$ are said to be scalar components.

EXAMPLE: Find the horizontal and vertical components of the force exerted on the bolt (See adjacent figure).

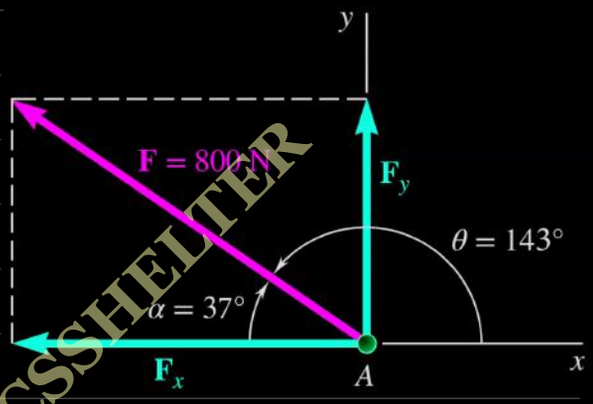


Solution: For purpose of finding components we enclose the force in a rectangle, keeping force as the diagonal as shown in adjacent figure. Now $F_x = F\cos(143°) = 800 \times \frac{-4}{5}$ N

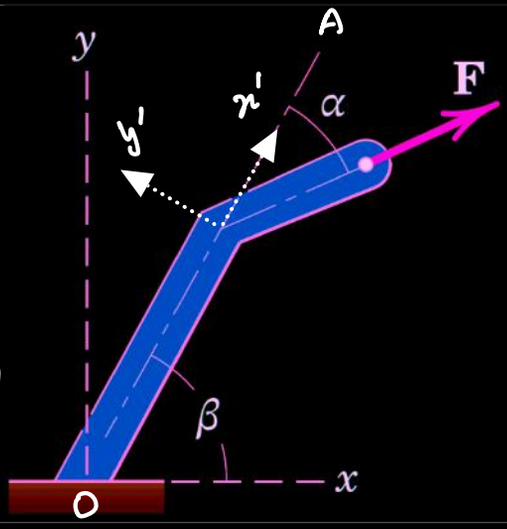
$\Rightarrow F_x = -640$ N similarly, $F_y = F\sin 143°$

$\Rightarrow F_y = F\sin 37° = 800 \times \frac{3}{5}$ N $= 480$ N.

Thus, $\vec{F} = -640\,\hat{i} + 480\,\hat{j}$ N. Ans.



EXAMPLE: For the adjacent figure, resolve the force F in to (a) components along x and y axes (b) components parallel and ⊥ to OA.



Solution: Angle of the force with +Ve x-axis is $\beta - \alpha$. Thus: $F_x = F\cos(\beta-\alpha)$, $F_y = F\sin(\beta-\alpha)$

(b) With line OA or x' axis the angle of the force is $-\alpha$. Thus:

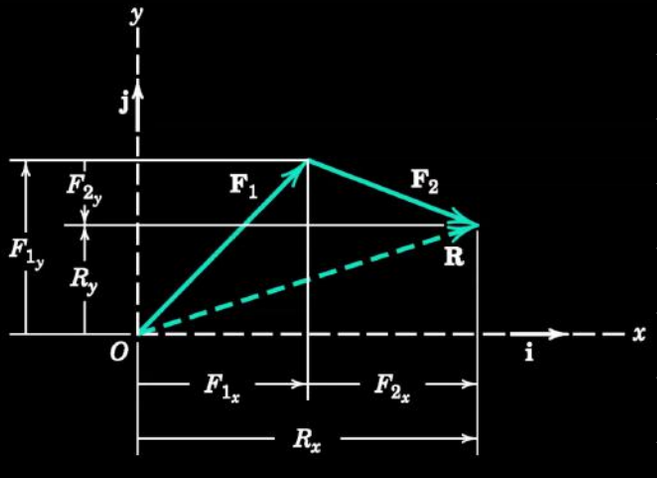
$F_{x'} = F\cos(-\alpha) = F\cos\alpha$, $F_{y'} = F\sin(-\alpha) = -F\sin\alpha$ Ans.

ADDITION OF VECTORS IN CARTESIAN FORM:

Consider the adjacent figure representing addition of forces $\vec{F_1}$ and $\vec{F_2}$. The resultant is $\vec{R}$. Now:

$\vec{R_y} = \vec{F_{1y}} + \vec{F_{2y}}$ and

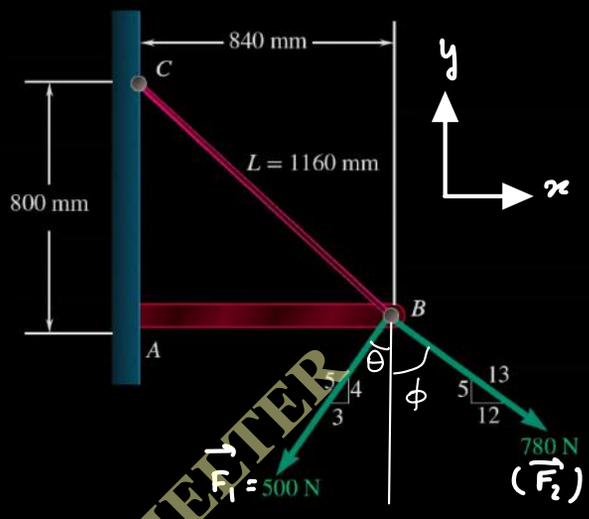
$\vec{R_x} = \vec{F_{1x}} + \vec{F_{2x}}$ thus we can see that for purpose of finding resultant



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of vectors expressed in cartesian form, their x components and y components can be added separately. Thus if $\vec{F_1} = 4\hat{i} + 4\hat{j}$ and $\vec{F_2} = 3\hat{i} - \hat{j}$ then $\vec{F_1} + \vec{F_2} = (4+3)\hat{i} + (4-1)\hat{j}$ i.e $\vec{R} = 7\hat{i} + 3\hat{j}$ Ans.

Example: Two forces of magnitudes 500 N and 780 N are acting at point B as shown. What is the resultant of the two forces?



Solution: $\vec{F} = \vec{F_1} + \vec{F_2}$

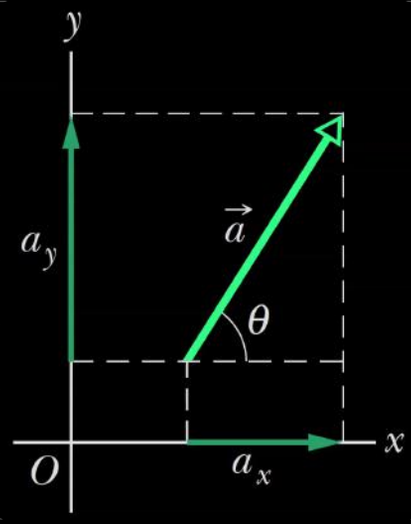
$= 500(-\sin\theta\,\hat{i} - \cos\theta\,\hat{j}) + 780(\sin\phi\,\hat{i} - \cos\phi\,\hat{j})$

$= 420\hat{i} - 700\hat{j}$ Ans.

$\sin\theta = 3/5,\ \cos\theta = 4/5,\ \sin\phi = 12/13$

$\cos\phi = 5/13.$

Polar Form Of A Vector: In a previous section we saw how to resolve a vector in to components along x and y axes and hence express the vector in form $\vec{a} = a_x\hat{i} + a_y\hat{j}$, alternatively a vector could also be uniquely specified by specifying its angle from x axis (say θ) and its magnitude and we could write the vector in shorthand notation as $\vec{a} = a\angle\theta$. This is known as polar form of expressing the vector. To convert a vector from cartesian to polar form consider the adjacent figure, we can readily observe that:



$$a = \sqrt{a_x^2 + a_y^2} \quad \text{and} \quad \tan\theta = a_y/a_x$$

Now recall that principle range of $\tan^{-1}$ is from $-90°$ to $+90°$ and a given value of $\tan\theta$ could also imply an angle in second or third quadrant as well, this happens when a_x is negative. This gives us following formula for finding θ:

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$\theta = \tan^{-1}(a_y/a_x)$ if $a_x > 0$ and $\theta = 180^\circ + \tan^{-1}(a_y/a_x)$ if $a_x < 0$

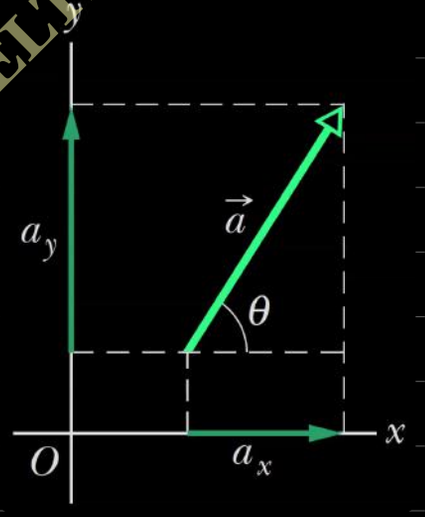
Example: Convert the following vectors from cartesian to polar form:
(a) $3\hat{i} + 4\hat{j}$ (b) $-3\hat{i} + 4\hat{j}$ (c) $3\hat{i} - 4\hat{j}$ (d) $-3\hat{i} - 4\hat{j}$

Solution: (a) $5\angle \tan^{-1}(4/3) \approx 5\angle 53^\circ$ (b) $5\angle(180^\circ + \tan^{-1}(-4/3)) = 5\angle 127^\circ$
(c) $5\angle \tan^{-1}(-4/3) = 5\angle -53^\circ$ (d) $5\angle[180^\circ + \tan^{-1}(4/3)] = 5\angle 233^\circ$

Converting From Polar To Cartesian Form:

Consider adjacent figure, from this one can readily see the polar to cartesian conversion as:
$a_x = a\cos\theta$ and $a_y = a\sin\theta$ thus:

$$\vec{a} = a\cos\theta\,\hat{i} + a\sin\theta\,\hat{j}$$

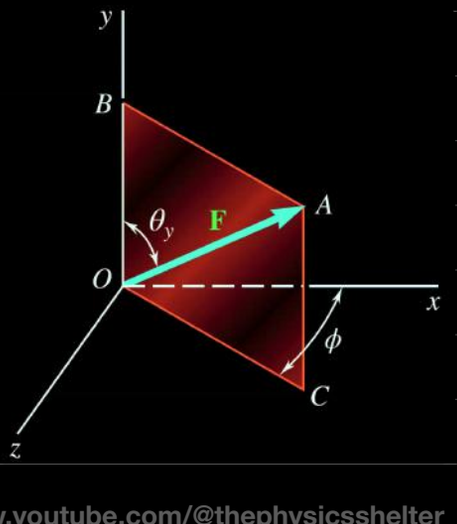


Example: Convert from polar to cartesian form:
(a) $10\angle 30^\circ$ (b) $10\angle 330^\circ$ (c) $10\angle 150^\circ$ (d) $10\angle 210^\circ$ (e) $10\angle -60^\circ$

Solution: (a) $10\cos 30^\circ\,\hat{i} + 10\sin 30^\circ\,\hat{j} = 5\sqrt{3}\,\hat{i} + 5\hat{j}$ (b) $10\cos(-30^\circ)\,\hat{i} + 10\sin(-30^\circ)\,\hat{j}$
$\Rightarrow \vec{a} = 5\sqrt{3}\,\hat{i} - 5\hat{j}$ (c) $10\cos 150^\circ\,\hat{i} + 10\sin 150^\circ\,\hat{j} = -5\sqrt{3}\,\hat{i} + 5\hat{j}$
(d) $10\cos 210^\circ\,\hat{i} + 10\sin 210^\circ\,\hat{j} = -5\sqrt{3}\,\hat{i} - 5\hat{j}$ (e) $10\cos(-60^\circ)\,\hat{i} + 10\sin(-60^\circ)\,\hat{j}$
$\Rightarrow \vec{a} = 5\hat{i} - 5\sqrt{3}\,\hat{j}$ Ans.

Vectors In 3 Dimensions (Z-Axis):

So far we were expressing all forces in terms of only x and y components. However x-y coordinates can only express the forces in the x-y plane, though a force may have a component perpendicular to x-y plane as well. (Consider adjacent figure). To efficiently express



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such vectors in Cartesian form, we define another unit vector along Z axis which is coming out of the plane of the page. We call this as unit vector $\hat{k}$. Now we can easily express the given vector using $\hat{i}$, $\hat{j}$ and $\hat{k}$ unit vectors. For instance the shown vector has a component $F\cos\theta_y$ along $\hat{j}$ and $F\sin\theta_y$ in the x-z plane. Now the component in x-z plane can be further resolved along x and z directions as: $F_x = F\sin\theta_y\cos\phi$ and $F_z = F\sin\theta_y\sin\phi$. Thus now:

$$\vec{F} = F\sin\theta_y\cos\phi\,\hat{i} + F\cos\theta_y\,\hat{j} + F\sin\theta_y\sin\phi\,\hat{k}$$

→ Note that $F = \sqrt{F_x^2 + F_y^2 + F_z^2}$, this can be easily shown as follows:

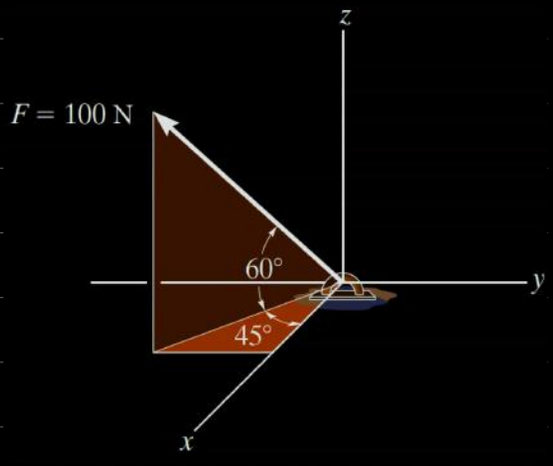
$$F_x^2 + F_z^2 = F^2\sin^2\theta_y\left[\sin^2\phi + \cos^2\phi\right] = F^2\sin^2\theta_y$$

Thus $F_x^2 + F_z^2 + F_y^2 = F^2\sin^2\theta_y + F^2\cos^2\theta_y = F^2$ Thus:

$$\boxed{F = \sqrt{F_x^2 + F_y^2 + F_z^2}}$$

→ This gives us a convenient way of finding magnitude of any 3D Vectors expressed in Cartesian form.

<u>EXAMPLE:</u> Express the 100 N force applied on the hook in terms of cartesian unit vectors.

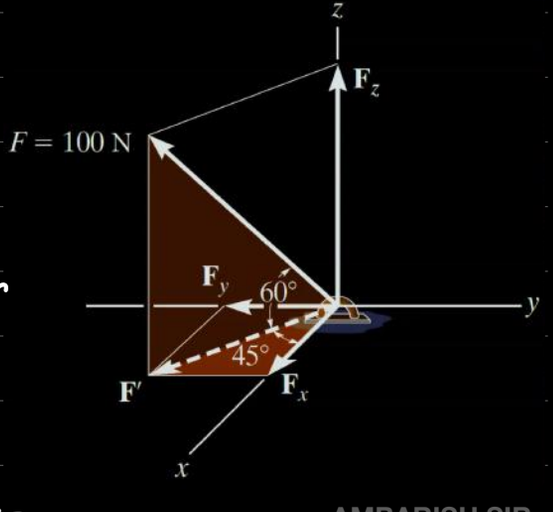


<u>Solution:</u> The force can be easily resolved by successive application of parallelogram law of addition (or repeatedly finding rectangular components). First we will resolve F in to components along Z axis and another in the x-y plane. Now the component in the x-y plane can be further resolved in to x and y components. Thus:

$$\vec{F} = 100\sin 60^\circ\,\hat{k} + 100\cos 60^\circ\cos 45^\circ\,\hat{i} - 100\cos 60^\circ\sin 45^\circ\,\hat{j}$$

Thus:

$$\vec{F} \approx 35.4\,\hat{i} - 35.4\,\hat{j} + 86.6\,\hat{k}\ \text{N}.$$

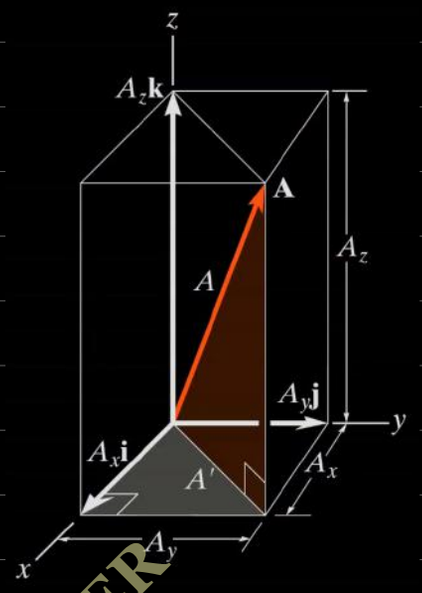


One can readily check $F = \sqrt{35.4^2 + 35.4^2 + 86.6^2} \approx 100\,\text{N}$.

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Visualizing 3-D Components As Sides Of Cuboid:

Just as in 2D plane, for purpose of resolving, we enclosed the vector in a rectangle, similarly, for a 3D vector, we can imagine enclosing the vector in a cuboid. For instance consider the adjacent figure, here A_x, A_y and A_z are simply sides of the cuboid whose body diagonal is vector $\vec{A}$.

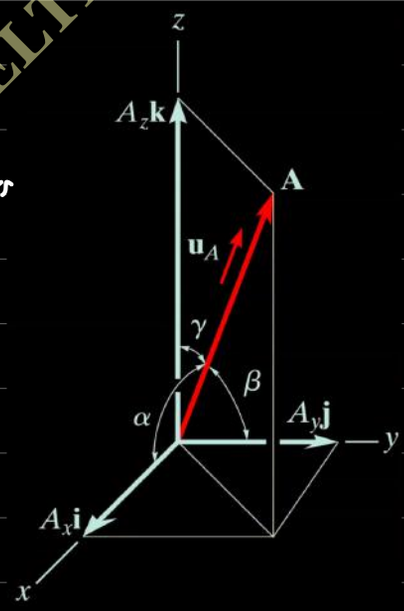


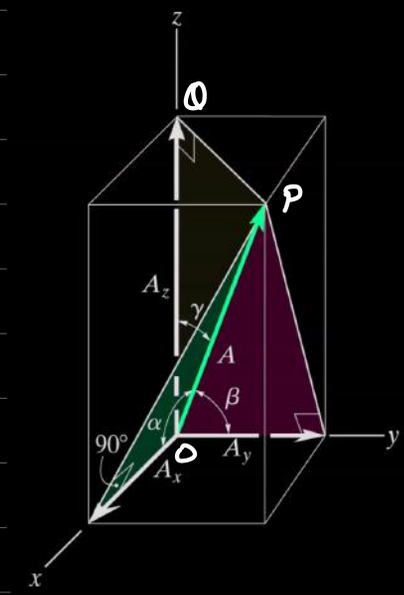
→ From here also it is easy to see that $A^2 = A_x^2 + A_y^2 + A_z^2$.

Direction Cosines Of A Vector:

Cosines of angles made by a vector with x, y and z axes are called direction cosines. Consider the adjacent figure, here x, y and z directions cosines are $\cos\alpha$, $\cos\beta$ and $\cos\gamma$.

We shall evaluate direction cosines in terms of components A_x, A_y and A_z. For this purpose, we first drop perpendiculars from head of vector $\vec{A}$ on x, y and z axes. These are shown in the next figure. (Visualize the figure carefully, Since we are seeing a 2D projection of a 3D figure, right angles may not appear to be 90° visually). Suppose we wish to now evaluate $\cos\gamma$, for this consider the right triangle OPQ. From this: $\cos\gamma = A_z/A$, Similarly $\cos\alpha = A_x/A$ and $\cos\beta = A_y/A$. Thus:





$$\cos\alpha = A_x/A \qquad \cos\beta = A_y/A \qquad \cos\gamma = A_z/A$$

→ Recall that unit vector along any vector can be obtained by dividing the vector by its magnitude i.e:

$$\hat{A} = \frac{\vec{A}}{A} = \frac{A_x\hat{i} + A_y\hat{j} + A_z\hat{k}}{A} = \cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k}$$

Thus unit vector along $\vec{A}$ is evaluated as $\hat{A} = \cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k}$

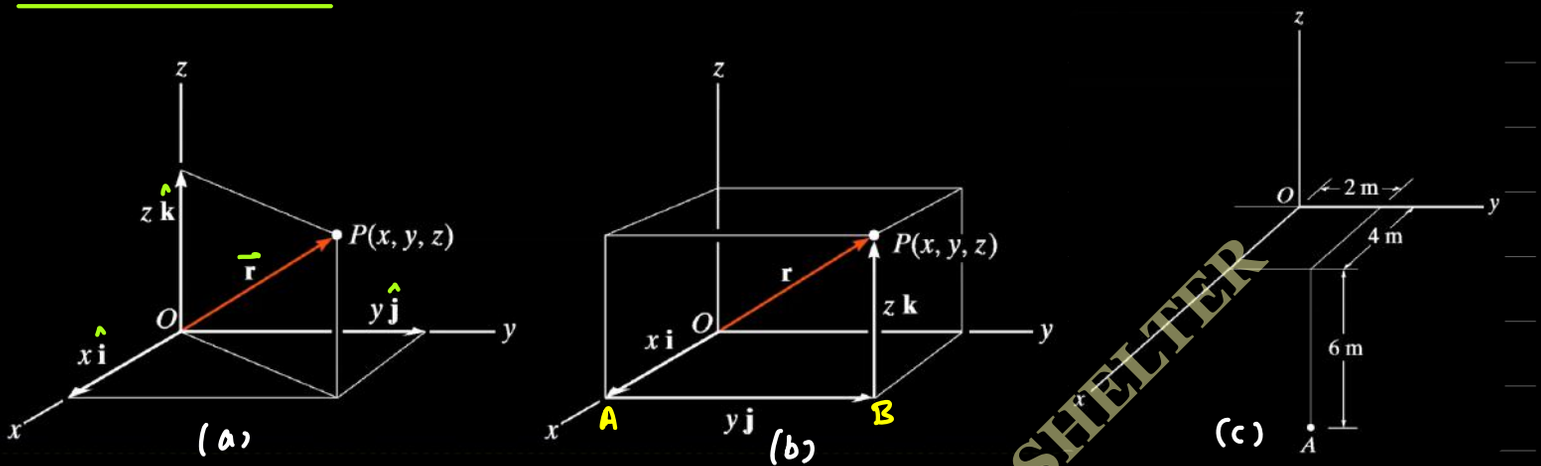
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→ One can readily verify that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ irrespective of the vector

Think: What will be $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$?

Ans: 2 ($\because \sin^2\alpha = 1 - \cos^2\alpha$ etc.).

Position Vector:

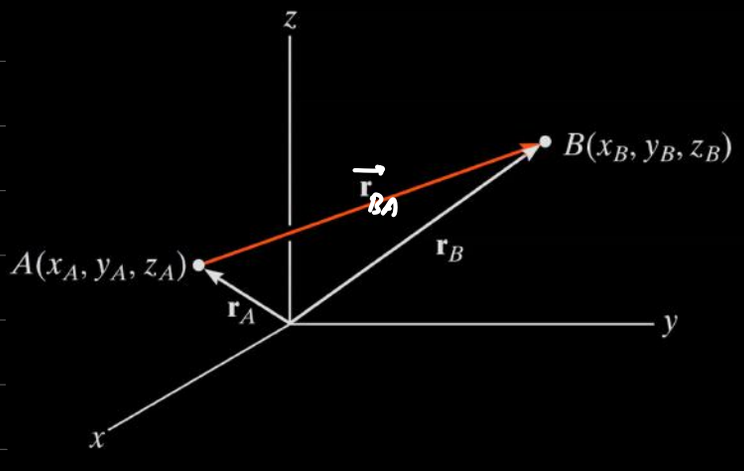


A vector joining the origin of coordinate system to the location of a point in space is called its position vector. Consider a vector $\vec{r}$ expressed as $x\hat{i} + y\hat{j} + z\hat{k}$ (This is nothing but vector sum $\vec{OA} + \vec{AB} + \vec{BP}$ (figure (b))), Here x, y and z are called coordinates of the point P. From figure (b) it is geometrically evident that coordinates simply tell us how much we have to move parallel to x, y and z axes to reach a particular point. For an antiparallel movement, a negative coordinate value is obtained. Thus if a point is located at (x, y, z) then its position vector is simply $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. For example in the third figure, position vector is simply $\vec{r} = 4\hat{i} + 2\hat{j} - 6\hat{k}$ m.

Relative Position Vector:

Let A and B be two points in space having position vectors $\vec{r}_A$ and $\vec{r}_B$, We define position vector of B relative to A ($\vec{r}_{BA}$) as a vector from point A to point B. Now using triangle law of addition, $\vec{r}_B = \vec{r}_A + \vec{r}_{BA}$, thus relative position vector of B wrt A:

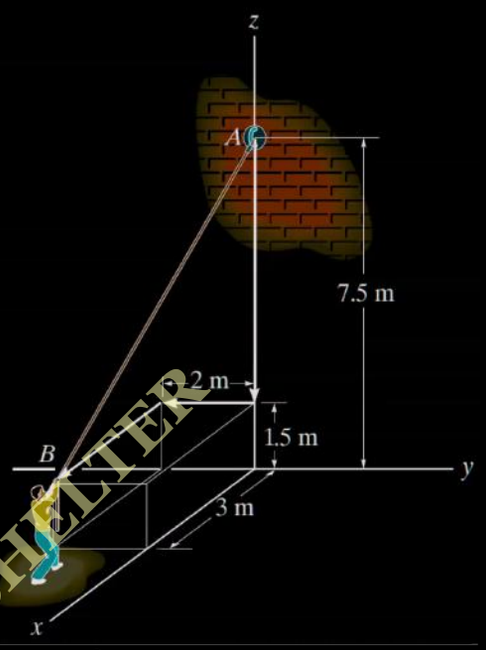
$$\vec{r}_{BA} = \vec{r}_B - \vec{r}_A$$



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Thus $\boxed{\vec{r}_{BA} = (x_B - x_A)\,\hat{i} + (y_B - y_A)\,\hat{j} + (z_B - z_A)\,\hat{k}}$

Example: A man pulls a cord with a force 350 N as shown in the figure. Express the force in cartesian coordinates.



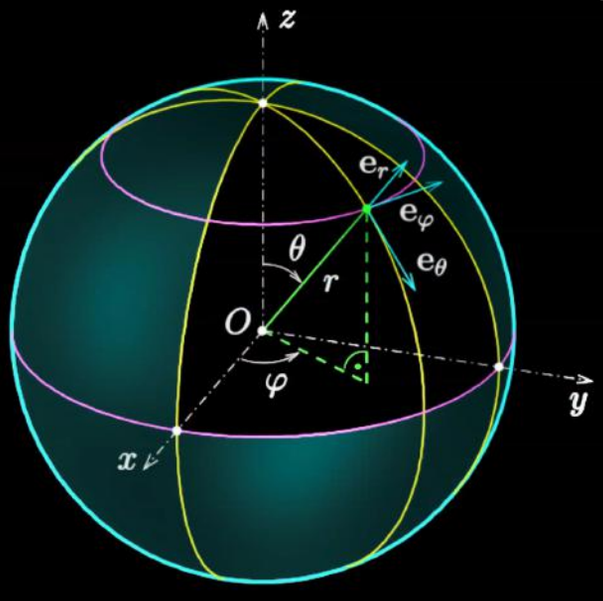
Solution: The force acts along $\vec{r}_{BA}$ thus required force $\vec{F} = F\hat{r}_{BA}$, so first we need to find $\vec{r}_{BA}$ and hence $\hat{r}_{BA}$. Now to reach B from A we have to move : 3m ll to x, 2m antiparallel to y and 6 m antiparallel to z axis thus, $\vec{r}_{BA} = 3\hat{i} - 2\hat{j} - 6\hat{k}$ m

Now $r_{BA} = \sqrt{3^2 + 2^2 + 6^2} = 7$ m Thus, $\hat{r}_{BA} = \dfrac{\vec{r}_{BA}}{r_{BA}} = \dfrac{3}{7}\hat{i} - \dfrac{2}{7}\hat{j} - \dfrac{6}{7}\hat{k}$

Now $\vec{F} = F\hat{F} = F\hat{r}_{BA} = 350\left(\dfrac{3}{7}\hat{i} - \dfrac{2}{7}\hat{j} - \dfrac{6}{7}\hat{k}\right)$ N $= 150\hat{i} - 100\hat{j} - 300\hat{k}$ N

Ans.

Spherical Coordinate System: To specify position of a coordinate in space three numerical value are required. In the cartesian coordinate system all the three values were distances however we could also specify the position using one distance and two angles, i.e (r, θ, ϕ). Consider adjacent figure, here any point could be located based on its angle θ (co-latitude) measured from "North" pole of the sphere, its angle in "equatorial plane" on



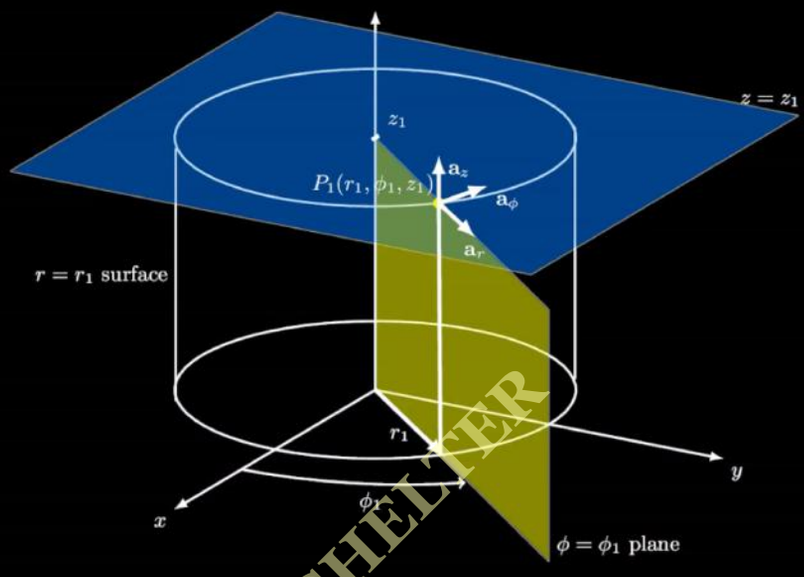
sent of a longitude angle ϕ (known as Azimuthal angle) and its distance from the origin. Thus using the set (r, θ, ϕ) the point is precisely located. Spherical coordinates are often convenient for analysis in astrophysics as well as atomic physics (such as figuring out the geometry of electron cloud around nucleus). Also it can

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be convenient in robotics for analysis and design of certain types of robots, and missile / Ballistic systems in military applications.

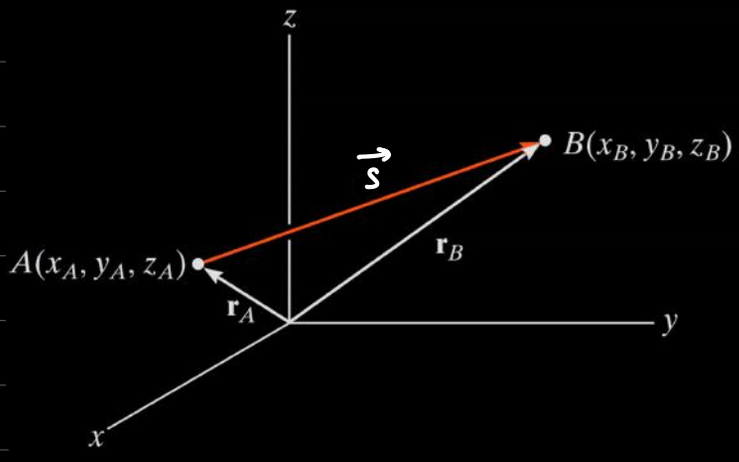
Cylindrical Coordinate System:

An alternate way of specifying position of a point in space involves specifying one angular coordinate and two distances. For instance consider the adjacent figure. Here coordinates of point P_1 are expressed in terms of its azimuthal angle (ϕ), radius (r) and altitude (or height) h. Thus the set (r, ϕ, z) completely specifies the points. (This is also intersection of three surfaces shown in the figure).



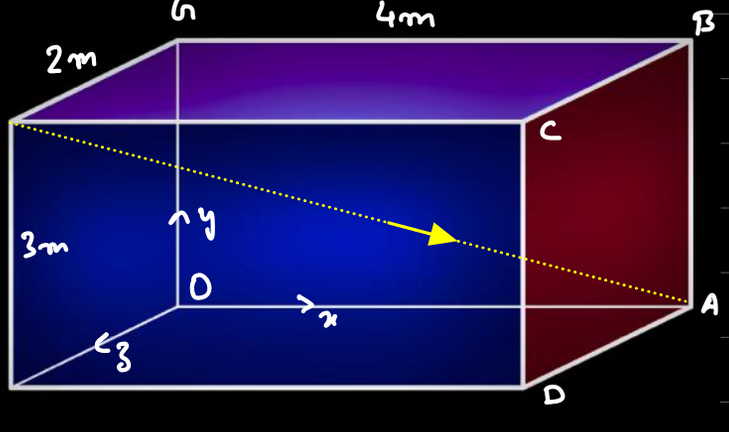
Displacement Vector:

If a particle is moving in space, the change in its position vector is called displacement vector. For instance if a particle moves from position A to position B, its displacement vector is given as $\vec{s} = \overrightarrow{AB} = \vec{r}_B - \vec{r}_A$ this is also same as position vector joining initial position to final position.



Example:

A mosquito flies from point F to A inside a mosquito net calculate (a) its displacement vector (b) Unit vector and direction cosines along displacement.



Solution: To reach from F to A, one has to move 4 units || to x axis,

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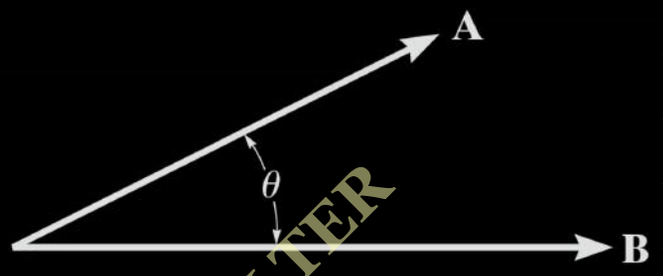
3 units ↓↑ to y axis and 2 units ↑↓ to z axis thus required displacement vector is $\vec{R}_{BA} = 4\hat{i} - 3\hat{j} - 2\hat{k}$ m. Now $r_{AB} = \sqrt{29}$ m

Thus $\hat{r}_{AB} = \frac{4}{\sqrt{29}}\hat{i} - \frac{3}{\sqrt{29}}\hat{j} - \frac{2}{\sqrt{29}}\hat{k}$ and the direction cosines, $\cos\alpha = 4/\sqrt{29}$,

$\cos\beta = -3/\sqrt{29}$, and $\cos\gamma = -2/\sqrt{29}$.

Dot Product: Let $\vec{A}$ and $\vec{B}$ be two vectors inclined at an angle θ to each other, then we define their dot product:

$$\vec{A}\cdot\vec{B} = AB\cos\theta.$$



→ Recall that θ is always considered as non-reflex angle between vectors when they are joined tail to tail.

→ Dot product is commutative i.e $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}$

→ Dot product is distributive i.e. $\vec{A}\cdot(\vec{B}+\vec{D}) = \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{D}$

→ Multiplication by a scalar: $a(\vec{A}\cdot\vec{B}) = (a\vec{A})\cdot\vec{B} = \vec{A}\cdot(a\vec{B})$

Dot Product Of Vectors Expressed In Cartesian Form:

$$\begin{aligned}\vec{\mathbf{A}}\cdot\vec{\mathbf{B}} &= (A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}})\cdot(B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}}) \\ &= A_xB_x(\hat{\mathbf{i}}\cdot\hat{\mathbf{i}}) + A_xB_y(\hat{\mathbf{i}}\cdot\hat{\mathbf{j}}) + A_xB_z(\hat{\mathbf{i}}\cdot\hat{\mathbf{k}}) \\ &+ A_yB_x(\hat{\mathbf{j}}\cdot\hat{\mathbf{i}}) + A_yB_y(\hat{\mathbf{j}}\cdot\hat{\mathbf{j}}) + A_yB_z(\hat{\mathbf{j}}\cdot\hat{\mathbf{k}}) \\ &+ A_zB_x(\hat{\mathbf{k}}\cdot\hat{\mathbf{i}}) + A_zB_y(\hat{\mathbf{k}}\cdot\hat{\mathbf{j}}) + A_zB_z(\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}) \quad (1)\end{aligned}$$

Now, $\hat{i}\cdot\hat{i} = 1\times1\times\cos 0^\circ = 1$ $\quad \hat{i}\cdot\hat{j} = 1\times1\times\cos 90^\circ = 0$ Thus:

$\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$ and $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$, Now carrying out all the dot products in equation (1):

$$\boxed{\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z}$$

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FINDING ANGLE BETWEEN TWO VECTORS USING DOT PRODUCT:

$$\vec{A}\cdot\vec{B} = AB\cos\theta$$

$$\Rightarrow \boxed{\cos\theta = \left(\frac{\vec{A}\cdot\vec{B}}{AB}\right) = \hat{A}\cdot\hat{B}}$$

→ This formula can be used for quickly finding angles between any two vectors expressed in cartesian form.

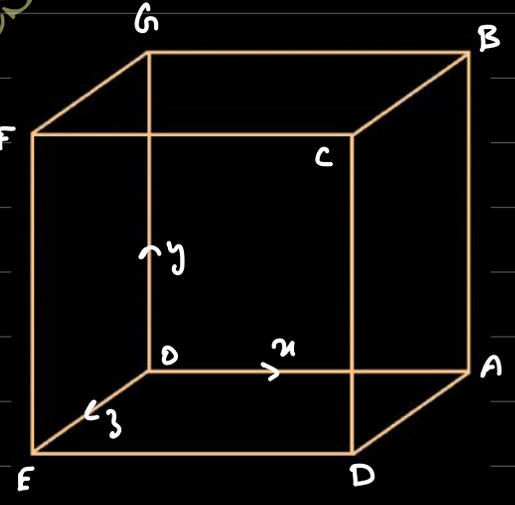
→ For mutually perpendicular vectors, $\vec{A}\cdot\vec{B} = 0$ ∵ $\cos 90° = 0$.

EXAMPLE: Find angle between vectors $\vec{a} = 3\hat{i} - 4\hat{j}$ and $\vec{b} = 2\hat{i} - \hat{j}$.

Solution: $a = \sqrt{3^2+4^2} = 5$, $b = \sqrt{2^2+1^2} = \sqrt{5}$, $\vec{a}\cdot\vec{b} = 6+4 = 10$

Thus $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{ab} = \frac{10}{5\sqrt{5}} = \frac{2}{\sqrt{5}} \Rightarrow \theta = \cos^{-1}(2/\sqrt{5})$ Ans.

EXAMPLE: Consider a unit cube shown in the figure. Find angle between : (a) $\vec{FD}$ and $\vec{FB}$ (b) $\vec{FA}$ and $\vec{FC}$



Solution: (a) $\vec{FD} = \hat{i} - \hat{j} \Rightarrow FD = \sqrt{2}$, $\vec{FB} = \hat{i} - \hat{k}$

$\Rightarrow FB = \sqrt{2}$ Now $\cos\theta = \vec{FD}\cdot\vec{FB}/FD\times FB$

$\Rightarrow \cos\theta = 1/(\sqrt{2}\times\sqrt{2}) = 1/2 \Rightarrow \theta = 60°$ Ans.

(b) $\vec{FA} = \hat{i} - \hat{j} - \hat{k} \Rightarrow FA = \sqrt{3}$, $\vec{FC} = \hat{i} \Rightarrow FC = 1$

Now $\cos\theta = \frac{\vec{FA}\cdot\vec{FC}}{FA\times FC} = \frac{1}{\sqrt{3}} \Rightarrow \theta = \cos^{-1}(1/\sqrt{3})$

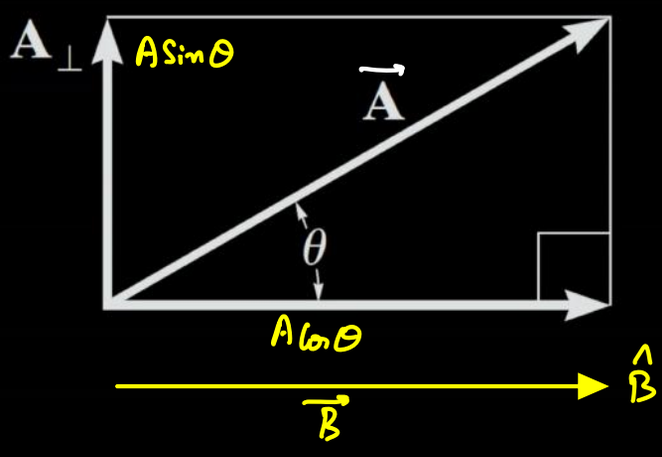
COMPONENTS OF A VECTOR ∥ AND ⊥ TO A GIVEN DIRECTION USING DOT PRODUCT:

Suppose we wish to resolve a vector $\vec{A}$ in to components parallel and perpendicular to another vector then the parallel component:

$$A_{\parallel} = A\cos\theta = \vec{A}\cdot\hat{B}$$

Thus $\boxed{A_{\parallel} = \vec{A}\cdot\hat{B} = \frac{\vec{A}\cdot\vec{B}}{B}}$

Vectorially $\boxed{\vec{A}_{\parallel} = (\vec{A}\cdot\hat{B})\hat{B}}$

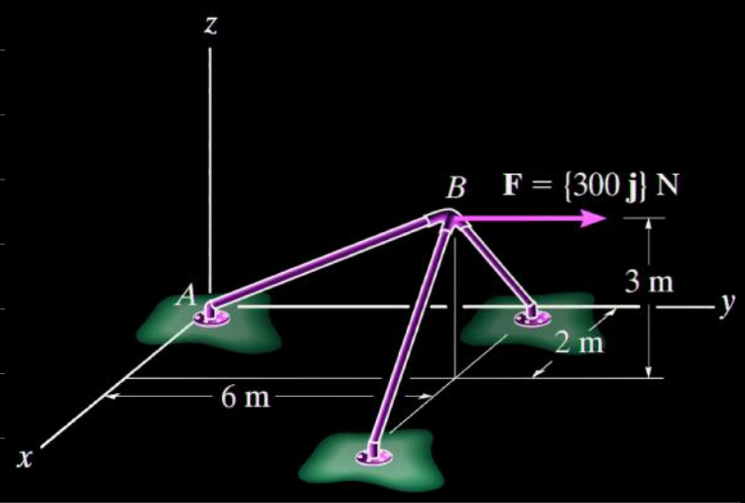


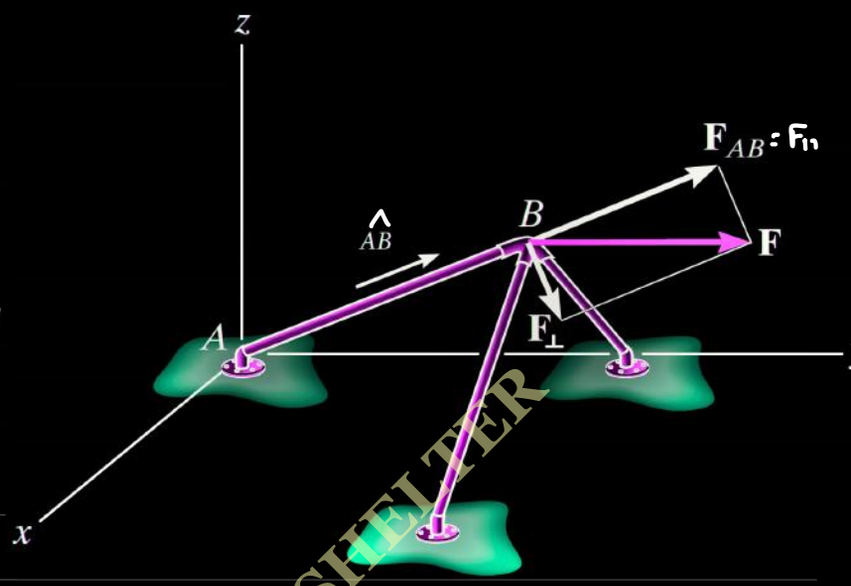
Now $\vec{A}_{\parallel} + \vec{A}_{\perp} = \vec{A} \Rightarrow \vec{A}_{\perp} = \vec{A} - \vec{A}_{\parallel} = \vec{A} - (\vec{A}\cdot\hat{B})\hat{B}$

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Thus $\boxed{\vec{A}_\perp = \vec{A} - (\vec{A}\cdot\hat{B})\hat{B}}$

EXAMPLE: Find the components of $\vec{F}$ parallel and perpendicular to rod AB.



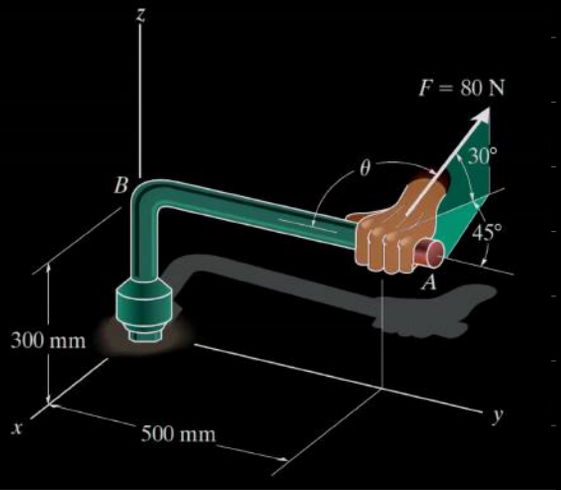


Solution: $\vec{AB} = 2\hat{i} + 6\hat{j} + 3\hat{k} \Rightarrow AB = \sqrt{2^2+6^2+3^2} = 7\ m$

Now $\vec{F}_{\parallel} = \left(\vec{F}\cdot\frac{\vec{AB}}{AB}\right)\hat{AB} = \left(\frac{1800}{7}\right)\left[\frac{2\hat{i}+6\hat{j}+3\hat{k}}{7}\right] \approx 73.5\hat{i} + 220\hat{j} + 110\hat{k}\ N$

Now $\vec{F}_\perp = \vec{F} - \vec{F}_{\parallel} \approx 300\hat{j} - [73.5\hat{i} + 220\hat{j} + 110\hat{k}] = -73.5\hat{i} + 79.6\hat{j} - 110\hat{k}$. Ans.

EXAMPLE: A force F = 80N is applied to the handle of the wrench. Determine the angle between the tail of the force and the handle AB.



Solution: $\vec{F} = 80(\cos 30^\circ \sin 45^\circ)(-\hat{i}) + 80(\cos 30^\circ \cos 45^\circ)\hat{j} + 80\sin 30^\circ \hat{k} = -40\sqrt{\frac{3}{2}}\hat{i} + 40\sqrt{\frac{3}{2}}\hat{j} + 40\hat{k}$

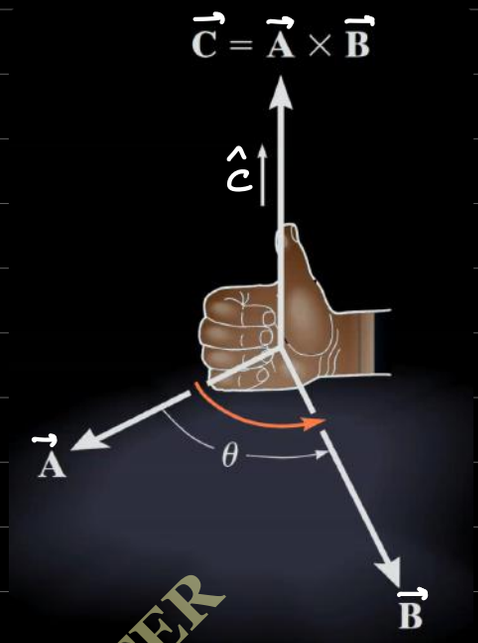
$\Rightarrow \vec{F} = -20\sqrt{6}\,\hat{i} + 20\sqrt{6}\,\hat{j} + 40\hat{k}$

Now unit vector along AB is $-\hat{j}$, Now

$\cos\theta = \frac{\vec{F}\cdot(-\hat{j})}{F\times 1} = \frac{-20\sqrt{6}}{80} \Rightarrow \theta = \cos^{-1}(\sqrt{6}/4)$ Ans.

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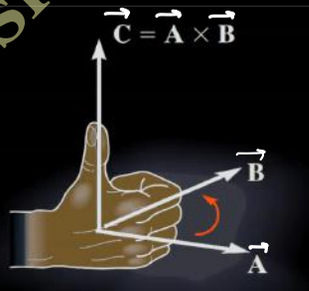
Cross Product: Let $\vec{A}$ and $\vec{B}$ be two vectors having an angle θ between them then we define their cross product $\vec{C} = \vec{A} \times \vec{B}$ as a vector with:



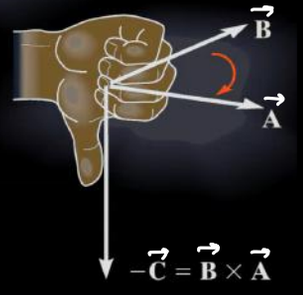
① Magnitude given as $C = AB \sin\theta$

② Direction given by right hand thumb rule: Align the little fingers of right hand on with vector A thumb ⊥ plane containing vector $\vec{A}$ and $\vec{B}$ such that curl of fingers points towards vector $\vec{B}$ through a non-reflex angle. (See figure).

→ Cross product is not commutative i.e.: $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$ rather $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$



See adjacent figure. Recall that while doing cross product curl of fingers has to point from first vector to the second vector with little finger aligned with first vector. Therefore when first vector and second vector are swapped, the thumb has to point in opposite direction.



→ If a cross product is multiplied by a scalar, it follows associative law i.e.:

$$a(\vec{A} \times \vec{B}) = \vec{A} \times a\vec{B} = (\vec{A} \times \vec{B})a$$

→ Cross product also follows distributive law i.e.:

$$\vec{A} \times (\vec{B} + \vec{D}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{D}$$

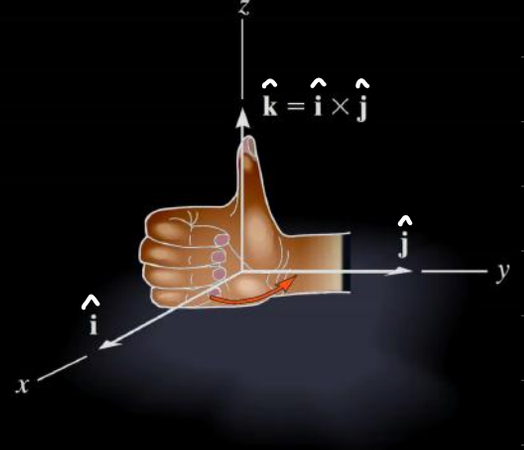
(Proper order of cross product should be maintained since it is not commutative).

Cross Product In Cartesian Form:

$$\hat{i} \times \hat{j} = \hat{k} \qquad \hat{i} \times \hat{k} = -\hat{j} \qquad \hat{i} \times \hat{i} = 0$$

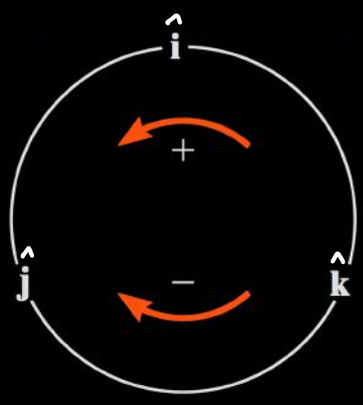
$$\hat{j} \times \hat{k} = \hat{i} \qquad \hat{j} \times \hat{i} = -\hat{k} \qquad \hat{j} \times \hat{j} = 0$$

$$\hat{k} \times \hat{i} = \hat{j} \qquad \hat{k} \times \hat{j} = -\hat{i} \qquad \hat{k} \times \hat{k} = 0$$



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The table above lists pairwise cross products of unit vectors along x, y and z axes. These results can also be easily memorized using the mnemonic in the adjacent figure. [$\hat{i} \times \hat{j} = \hat{k}$ ∵ anticlockwise, $\hat{j} \times \hat{i} = -\hat{k}$ ∵ clockwise, see arrows in figure].



Now let us consider cross product of two vectors $\vec{A}$ and $\vec{B}$:

$$\begin{aligned}\vec{\mathbf{A}} \times \vec{\mathbf{B}} &= (A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}) \times (B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}}) \\ &= A_xB_x(\hat{\mathbf{i}} \times \hat{\mathbf{i}}) + A_xB_y(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) + A_xB_z(\hat{\mathbf{i}} \times \hat{\mathbf{k}}) \\ &\quad + A_yB_x(\hat{\mathbf{j}} \times \hat{\mathbf{i}}) + A_yB_y(\hat{\mathbf{j}} \times \hat{\mathbf{j}}) + A_yB_z(\hat{\mathbf{j}} \times \hat{\mathbf{k}}) \\ &\quad + A_zB_x(\hat{\mathbf{k}} \times \hat{\mathbf{i}}) + A_zB_y(\hat{\mathbf{k}} \times \hat{\mathbf{j}}) + A_zB_z(\hat{\mathbf{k}} \times \hat{\mathbf{k}})\end{aligned}$$

$$\Rightarrow \vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\,\hat{i} - (A_x B_z - A_z B_x)\,\hat{j} + (A_x B_y - A_y B_x)\,\hat{k}$$

This result can be conveniently expressed in determinant form as follows:

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

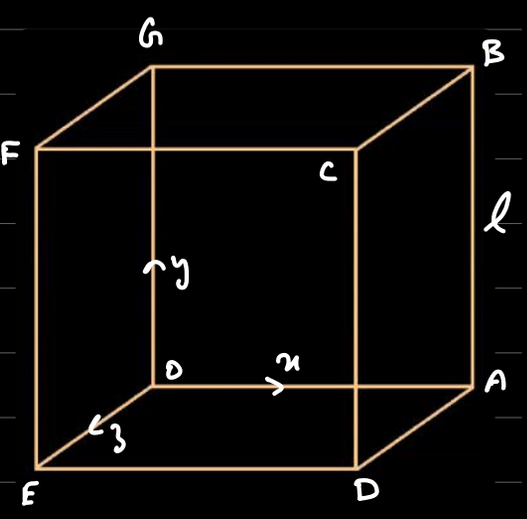
The determinant is expanded as per following scheme:

For element **i**: $\hat{\mathbf{i}}(A_yB_z - A_zB_y)$, For element **k**: $\hat{\mathbf{k}}(A_xB_y - A_yB_x)$

For element **j**: $-\hat{\mathbf{j}}(A_xB_z - A_zB_x)$ — Remember the negative sign

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Example: For the shown cube in adjacent figure, find the cross product $\vec{FA} \times \vec{FD}$, hence find a unit vector perpendicular to the plane containing $\vec{FA}$ and $\vec{FD}$.



Solution: $\vec{FA} = l(\hat{i} - \hat{j} - \hat{k})$, $\vec{FD} = l(\hat{i} - \hat{j})$

Thus $\vec{FA} \times \vec{FD}$ is obtained by the determinant:

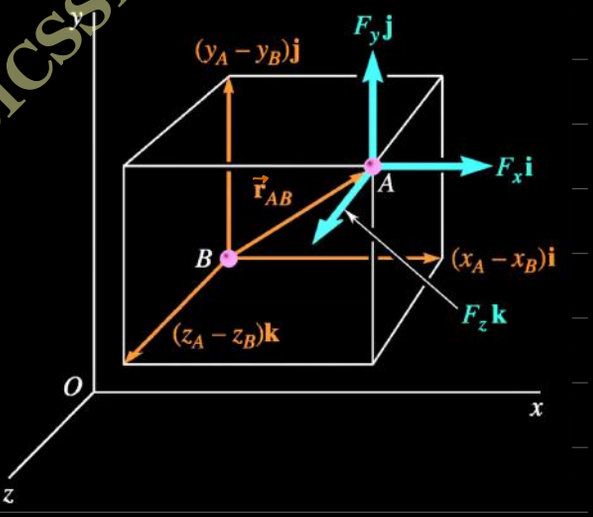
$$l^2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 1 & -1 & 0 \end{vmatrix} = (-\hat{i} - \hat{j}(1) - 2\hat{k})\, l^2 = (-\hat{i} - \hat{j} - 2\hat{k})\, l^2$$

Thus, $\vec{FA} \times \vec{FD} = (-\hat{i} - \hat{j} - 2\hat{k})\, l^2$, the required unit vector $\pm\left(\frac{\hat{i} + \hat{j} + 2\hat{k}}{\sqrt{6}}\right)$ Ans.

Torque (Moment) Of A Force:

Let a force $\vec{F}$ act at a point A whose position vector wrt a point B is $\vec{r}_{AB}$ then we define torque of the force about point B as:

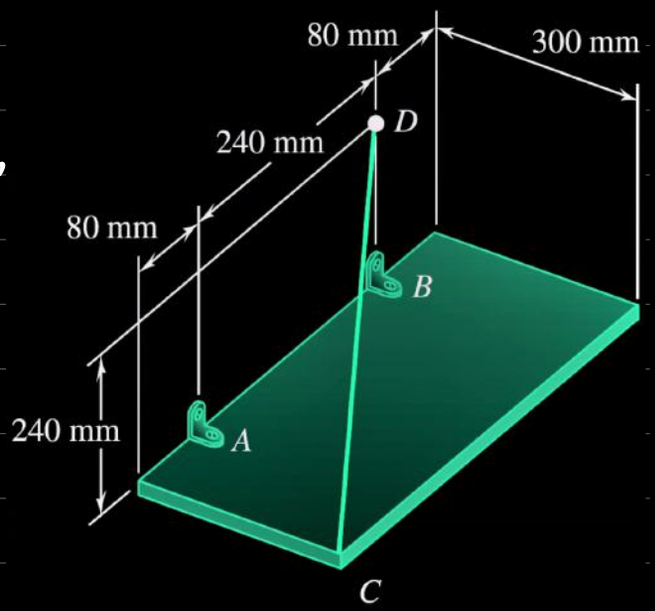
$$\boxed{\vec{\tau} = \vec{r}_{AB} \times \vec{F}}$$



→ Here $\vec{r}_{AB}$ is position of A wrt B i.e $\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = (x_A - x_B)\hat{i} + (y_A - y_B)\hat{j} + (z_A - z_B)\hat{k}$.

Example:

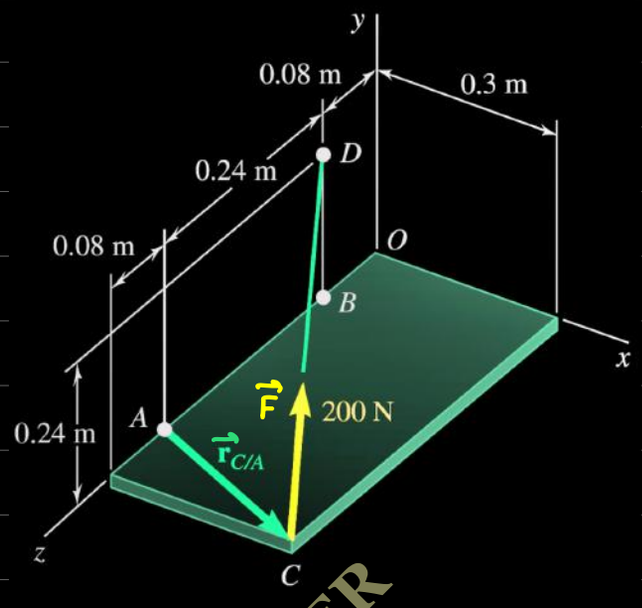
A rectangular plate is supported by brackets at A and B by a wire CD. If the tension in the wire is 200 N, determine the moment about A of the force exerted by the wire on point C.



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Solution: By definition, $\vec{\tau} = \vec{r}_{CA} \times \vec{F}$

Now $\vec{r}_{CA} = 0.3\hat{i} + 0.08\hat{k}$ m.



$\vec{F} = F\hat{CD}$, $\vec{CD} = -0.3\hat{i} + 0.24\hat{j} - 0.32\hat{k}$

$CD = \sqrt{0.3^2 + 0.24^2 + 0.32^2} = 0.5$ m.

Thus, $\vec{F} = \frac{200\text{ N}}{0.5\text{ m}}\left[-0.3\hat{i} + 0.24\hat{j} - 0.32\hat{k}\right]$

$\Rightarrow \vec{F} = (-120\hat{i} + 96\hat{j} - 128\hat{k})$ N, Thus required torque:

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.3 & 0 & 0.08 \\ -120 & 96 & -128 \end{vmatrix} \text{ Nm} = (-7.68\hat{i} + 28.8\hat{j} + 28.8\hat{k})\text{ Nm}$$

Ans.

Work Done By A Constant Force:



If a particle undergoes a displacement $\vec{s}$ while being acted upon by a constant force $\vec{F}$ then work done by the force is defined as $\boxed{W = \vec{F}\cdot\vec{s}}$

i.e $W = Fs\cos\theta$

→ Thus if a particle moves from point A to Point B while being acted upon by a force $\vec{F}$, $\vec{s} = \vec{r}_B - \vec{r}_A \Rightarrow W = \vec{F}\cdot(\vec{r}_B - \vec{r}_A)$.

Example: A body is displaced from $\vec{r}_A = (2\hat{i} + 4\hat{j} - 6\hat{k})$ m to $\vec{r}_B = (6\hat{i} - 4\hat{j} + 2\hat{k})$ m by a force $\vec{F} = 2\hat{i} + 3\hat{j} - \hat{k}$ N. Find the work done.

Solution: $W = \vec{F}\cdot\vec{s} = \vec{F}\cdot\Delta\vec{r}$, $\Delta\vec{r} = \vec{r}_B - \vec{r}_A = 4\hat{i} - 8\hat{j} + 8\hat{k}$, Thus

$W = (2\hat{i} + 3\hat{j} - \hat{k})\cdot(4\hat{i} - 8\hat{j} + 8\hat{k}) = -24$ Nm or -24 Joule.

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Test For Parallel Vectors:

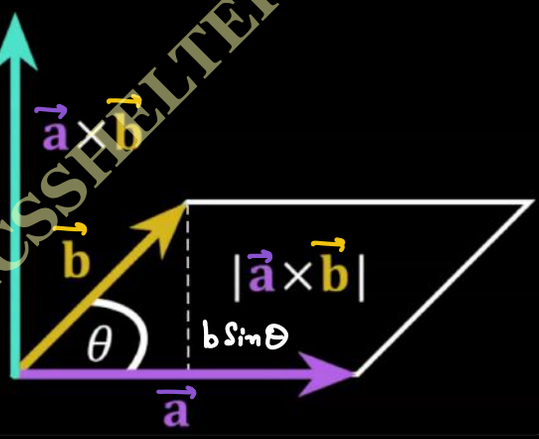
Two vectors $\vec{A}$ and $\vec{B}$ will be parallel if $\vec{B}$ can be expressed as some numerical multiple of $\vec{A}$ as $\vec{B} = \lambda\vec{A}$ where λ is some real number. Thus for parallel vectors, the coefficients should be in same ratio i.e:

$$\boxed{\frac{B_x}{A_x} = \frac{B_y}{A_y} = \frac{B_z}{A_z} = \lambda}$$

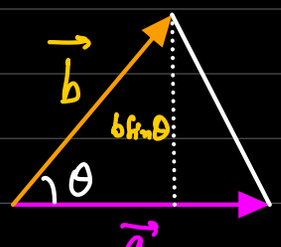
If λ is negative, vectors are considered antiparallel (a type of parallel).

Area Of A Parallelogram By Cross Product:

Consider a parallelogram defined by vectors $\vec{a}$ and $\vec{b}$ along its sides. The area of this parallelogram is $A = ab\sin\theta$ which can be evaluated as simply: $A = |\vec{a}\times\vec{b}|$. This can be convenient when finding area of a parallelogram defined in terms of vectors along its sides (being specified in $\hat{i}$, $\hat{j}$ and $\hat{k}$).



→ Similarly for a triangle defined by vectors $\vec{a}$ and $\vec{b}$ as its sides, the area is given as: $A = \frac{1}{2}ab\sin\theta = \frac{1}{2}|\vec{a}\times\vec{b}|$.



Scalar Triple Product:

Consider the parallelopipe defined by three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$

The volume:

V = Base Area × Height

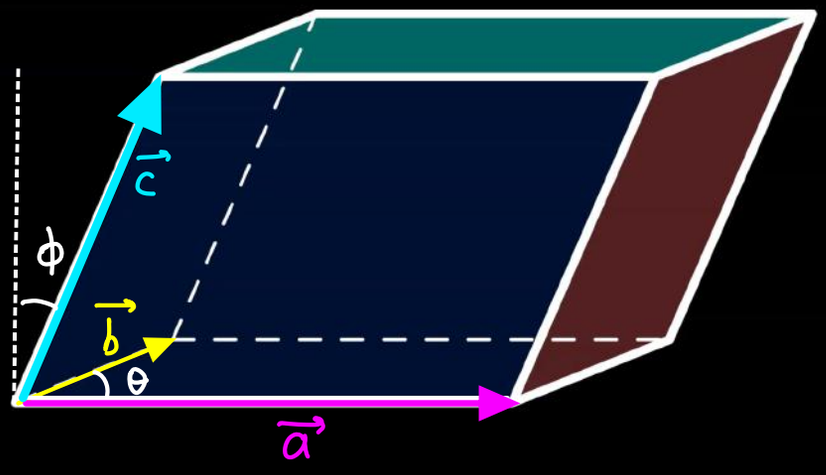
$= ab\sin\theta \times (\cos\phi$

$= |\vec{a}\times\vec{b}| \times \vec{c}\cdot(\hat{a}\times\hat{b})$

$= \vec{c}\cdot(\vec{a}\times\vec{b})$

Thus:

$$\boxed{V = |(\vec{a}\times\vec{b})\cdot\vec{c}|}$$

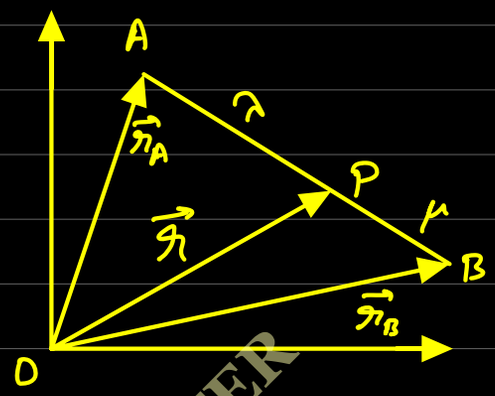


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Vector Triple Product:

$$\vec{a} \times [\vec{b} \times \vec{c}] = \vec{b}(\vec{a}\cdot\vec{c}) - \vec{c}(\vec{a}\cdot\vec{b})$$

Section Formula: We wish to locate position vector of a point P on line joining AB such that $AP : PB = \lambda : \mu$. We do this as follows:



$$\vec{AP} = \vec{r} - \vec{r}_A$$
$$\vec{PB} = \vec{r}_B - \vec{r}$$

Now $AP : PB = \lambda : \mu$, thus $\mu \vec{AP} = \lambda \vec{PB}$

$$\Rightarrow \mu(\vec{r} - \vec{r}_A) = \lambda(\vec{r}_B - \vec{r}) \Rightarrow (\lambda+\mu)\vec{r} = \lambda\vec{r}_B + \mu\vec{r}_A$$

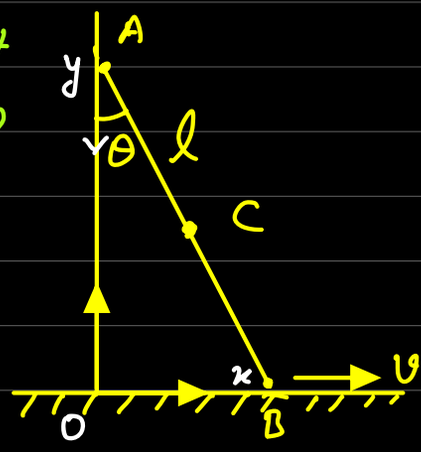
Thus $$\boxed{\vec{r} = (\lambda\vec{r}_B + \mu\vec{r}_A)/(\lambda+\mu)}$$ ← Section Formula.

→ If AB is a rigid body (or a part of a rigid body), section formula can be differentiated to get relations in velocities and accelerations. Thus:

$$\boxed{\vec{v}_p = (\lambda\vec{v}_B + \mu\vec{v}_A)/(\lambda+\mu)}$$

$$\boxed{\vec{a}_p = (\lambda\vec{a}_B + \mu\vec{a}_A)/(\lambda+\mu)}$$

Think: What will be the velocity vector of mid point of the sliding ladder shown in the figure? (B is being pulled with a speed v as shown)



Solution: $x^2 + y^2 = \ell^2 \Rightarrow 2x\dot{x} + y\dot{y} = 0$

$\Rightarrow \dot{y} = -\frac{x}{y}\dot{x} = -v\tan\theta$ thus:

$\vec{v}_A = -v\tan\theta\,\hat{j}$ $\quad \vec{v}_B = v\hat{i}$ Now by section formula,

$\vec{v}_c = \frac{\vec{v}_A + \vec{v}_B}{2} = (v\hat{i} - v\tan\theta\hat{j})/2$ Ans.

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